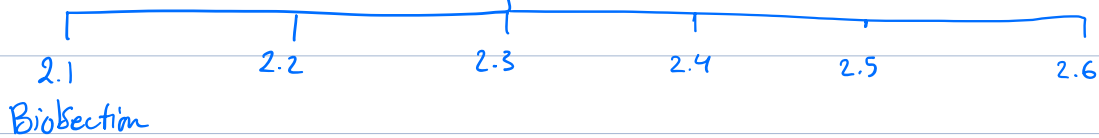


Chapter 2



The Bisection Method

Q1: Use the Bisection method to find p_3 for $f(x) = \sqrt{x} - \cos x = 0$ on $[0, 1]$.
 (Note: a is 0, b is 1. An arrow points to the equation with the text "should always $f(x) = 0$ ".)

① Check opposite signs:

$$f(a) = f(0) = \sqrt{0} - \cos(0) = -ve$$

$$f(b) = f(1) = \sqrt{1} - \cos(1) = +ve$$

② find p_1, p_2, p_3 :

$$\rightarrow p_n = \frac{a+b}{2}$$

$\rightarrow f(p_n) \Rightarrow$ to check sign and find next sequences

$\Rightarrow$

$$p_1 = \frac{0+1}{2} = \frac{1}{2} = \boxed{0.5}$$

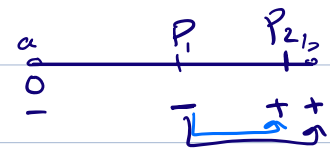
$$f(0.5) = \sqrt{0.5} - \cos(0.5) = -ve$$

* Set Calculator in Radian

$$p_2 = \frac{0.5+1}{2} = \boxed{0.75}$$

$$f(0.75) = +ve$$

$$p_3 = \frac{0.5+0.75}{2} = \boxed{0.625}$$



Q 2

Let $f(x) = 3(x+1)(x - \frac{1}{2})(x-1) = 0$. Use the Bisection method on the following intervals to find p_3 .

a. $\begin{matrix} a & b \\ [0, 1] \end{matrix}$

$$f(0) = +ve$$

$$f(1) = +ve$$

we can't continue solving the problem in same sign.

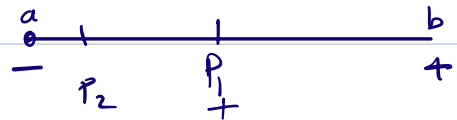
b. $\begin{matrix} a & b \\ [-1.25, 2.5] \end{matrix}$

$$f(a) = -ve$$

$$f(b) = +ve$$

$$p_1 = \frac{-1.25 + 2.5}{2} = \boxed{0.625}$$

$$f(0.625) = +ve$$



$$p_2 = \frac{-1.25 + 0.625}{2} = \boxed{-0.3125}$$

$$f(-0.3125) = -ve$$

$$p_3 = \frac{-0.3125 + 0.625}{2} = \boxed{0.15625}$$

Q3

Use the Bisection method to find solutions accurate to within 10^{-2} for $x^3 - 7x^2 + 14x - 6 = 0$ on each interval.

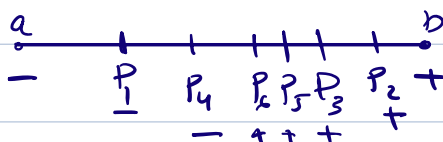
00

$$f(x) = x^3 - 7x^2 + 14x - 6$$

a. $\begin{matrix} a & b \\ [0, 1] \end{matrix}$

$$f(a) = -ve$$

$$f(b) = +ve$$



$$p_1 = \frac{0+1}{2} = 0.5$$

$$f(0.5) = -ve$$

$$|p_n - p| \leq 0.01$$

$$p_2 = \frac{0.5+1}{2} = 0.75$$

$$f(0.75) = +ve$$

$$\rightarrow |p_2 - p_1| = 0.25 \times$$

$$p_3 = \frac{0.5+0.75}{2} = 0.625$$

$$f(0.625) = +ve$$

$$\rightarrow |p_3 - p_2| = 0.125$$

$$p_4 = \frac{0.5+0.625}{2} = 0.5625$$

$$f(0.5625) = -ve$$

$$\rightarrow |p_4 - p_3| = 0.06$$

$$p_5 = \frac{0.5625+0.625}{2} = 0.59375$$

$$f(0.59375) = +ve$$

$$\rightarrow |p_5 - p_4| = 0.03$$

$$p_6 = \frac{0.5625+0.59375}{2} = 0.578125$$

$$f(0.578125) = +ve$$

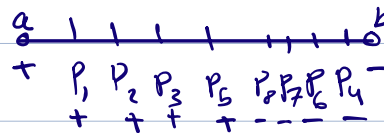
$$\rightarrow |p_6 - p_5| = 0.01$$

$$p_7 = \frac{0.5625 + 0.578125}{2} = 0.5703125 \rightarrow |p_7 - p_6| = 0.007$$

b. $\begin{matrix} a & b \\ [1, 3.2] \end{matrix}$

$$f(1) = +ve$$

$$f(3.2) = -ve$$



$$p_1 = \frac{1 + 3.2}{2} = 2.1$$

$$f(2.1) = +ve$$

$$p_2 = \frac{2.1 + 3.2}{2} = 2.65$$

$$f(2.65) = +ve$$

$$p_3 = \frac{2.65 + 3.2}{2} = 2.925$$

$$f(2.925) = +ve$$

$$p_4 = \frac{2.925 + 3.2}{2} = 3.0625$$

$$f(3.0625) = -ve$$

$$p_5 = \frac{2.925 + 3.0625}{2} = 2.99375$$

$$\rightarrow |p_5 - p_4| = > 10^{-2}$$

$$f(2.99375) = +ve$$

$$p_6 = \frac{2.99375 + 3.0625}{2} = 3.028125$$

$$\rightarrow |p_6 - p_5| = > 10^{-2}$$

$$f(p_6) = -ve$$

$$p_7 = \frac{2.99375 + 3.028125}{2} = 3.0109375$$

$$\rightarrow |p_7 - p_6| = > 10^{-2}$$

$$f(p_7) = -ve$$

$$P_8 = \frac{2.999375 + 3.0109375}{2} = 3.00234375 \rightarrow |P_8 - P_7| = 0.008 < 10^{-2}$$

$$f(P_8) = -ve$$

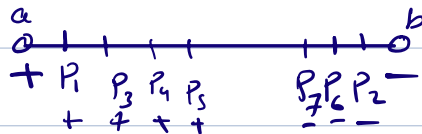
Q 4 Use the Bisection method to find solutions accurate to within 10^{-2} for $x^4 - 2x^3 - 4x^2 + 4x + 4 = 0$ on each interval.

$$f(x) = x^4 - 2x^3 - 4x^2 + 4x + 4$$

b. $[0, 2]$

$$f(a) = +ve$$

$$f(2) = -ve$$



$$P_1 = \frac{0+2}{2} = 1$$

$$f(P_1) = +ve$$

$$P_2 = \frac{1+2}{2} = 1.5$$

$$f(P_2) = -ve$$

$$P_3 = \frac{1+1.5}{2} = 1.25$$

$$f(P_3) = +ve$$

$$P_4 = \frac{1.25+1.5}{2} = 1.375$$

$$f(P_4) = +ve$$

$$P_5 = \frac{1.375 + 1.5}{2} = 1.4375$$

$$f(P_5) = +ve$$

$$P_6 = \frac{1.4375 + 1.5}{2} = 1.46875$$

$$f(P_6) = -ve$$

$$P_7 = \frac{1.4375 + 1.46875}{2} = 1.453125$$

$$f(P_7) = -ve^2$$

$$P_8 = \frac{1.4375 + 1.453125}{2} = \boxed{1.4453125} \rightarrow |P_8 - P_7| = 0.007$$